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Metal rib overmolding joining

Abstract

Metal rib overmolding joining including creating geometrical features in reinforcements for interconnecting a three-dimensional metal reinforcement to a metal base plate of an overmolding part in order to transfer load forces. Interlocked reinforcement is pre-assembled or assembled in an injection molding tool. Geometric features are positioned and joined creating a connection geometry that is overmolded. The joints created between the metal reinforcements are done without a welding process.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a National Stage of PCT International Application No. PCT/US2020/038777, filed Jun. 19, 2020 and claims the benefit of U.S. Provisional Patent Application No. 62/864,685, filed Jun. 21, 2019, U.S. Provisional Patent Application No. 62/864,691, filed Jun. 21, 2019, U.S. Provisional Patent Application No. 62/864,785, filed Jun. 21, 2019, U.S. Provisional Patent Application No. 62/864,878, filed Jun. 21, 2019 and U.S. Provisional Patent Application No. 62/864,981, filed Jun. 21, 2019. The disclosures of the above applications are incorporated herein by reference.

FIELD OF THE INVENTION

(1) The present invention relates to strengthening rib overmolding joining technologies.

BACKGROUND OF THE INVENTION

(2) Large panels, such as, but not limited to, liftgate panels for vehicles, generally require reinforcement in areas to meet structural performance and standards. A conventional reinforcement is a metal bracket. Standard flat metal bracket overmolding does not fulfill the customer requirements regarding structural performance. In addition, flat metal to flat metal bracket (flat-to-flat) has creep and lower stiffness. Stamping steel or other metal is also prone to waste of material, e.g., such as for flat-to-flat before further processing, and it is heavier, and requires thicker pieces of metal. Known hollow tubes for stiffening, e.g., metal tube, collapse during processing.

Furthermore, welding metal plates, such as flat metal plates, is expensive

(3) Steel, aluminum and magnesium castings fulfill the same performance requirements, but are much more expensive, higher weight, or have to be installed with secondary operations. Conventional liftgate panels, for example, are large and made of steel, aluminum and magnesium castings that are expensive, higher weight, and have to be installed with secondary operations.

(4) Accordingly, there is a need for a three dimensional structure with overmolding that gives significant performance benefits, including, load transfer by overmolding the connection, and method for manufacturing same that creates joints between metal parts without a welding process is a significant benefit.

SUMMARY OF THE INVENTION

(5) Three dimensional metal ribs need to be connected to a metal base plate of overmolding part in order to transfer load, which would require welding. However, aspects of the present invention utilize the molding process to connect and position the ribs to the base metal plate. Creating geometrical features in the base plate and the rib accommodate the above joining. Assembly and joining is happening before the overmolding or is happening in the injection tool.

(6) The present invention provides structural benefits to large panels due to geometrical features in the base plate and at least one rib in combination with the benefit of the overmolding process. Utilize the molding process to connect and position the ribs to the base metal plate. Create geometrical features in the base plate and the rib to accommodate the above. Assembly and joining is happening before the overmolding or is happening in the injection tool. Creating joints between metal parts without a welding process is a significant benefit. Load transfer by overmolding connection.

(7) Further areas of applicability of the present invention will become apparent from the detailed description provided hereinafter. It should be understood that the detailed description and specific examples, while indicating the preferred embodiment of the invention, are intended for purposes of illustration only and are not intended to limit the scope of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The present invention will become more fully understood from the detailed description and the accompanying drawings, wherein:
- (2) FIG. 1 is a perspective view of a joint formed by interlocking metal parts with an overmold, in accordance with the present invention;
- (3) FIG. 2 is a cross sectional view of the metal rib overmolding, according to aspects of the present invention;
- (4) FIG. 3 is a perspective view of a joint formed by interlocking metal parts with an overmold, in accordance with the present invention;
- (5) FIG. 4 is a cross sectional view taken at 4-4 of FIG. 3, according to aspects of the present invention;
- (6) FIG. 5 is a perspective view of a joint formed by interlocking metal parts with an overmold, in accordance with the present invention;
- (7) FIG. 6 is a cross sectional view of FIG. 5, according to aspects of the present invention;
- (8) FIG. 7 is a cross sectional view taken at 7-7 of FIG. 6;
- (9) FIG. 8 is a perspective view of a joint formed by interlocking metal parts with an overmold, in accordance with the present invention;
- (10) FIG. 9 is a perspective view of a joint formed by interlocking metal parts with an overmold, in accordance with the present invention;
- (11) FIG. 10 is a perspective view of a joint formed by interlocking metal parts with an overmold, in accordance with the present invention;
- (12) FIG. 11 is a perspective view of a joint formed by interlocking metal parts with an overmold, in accordance with the present invention;
- (13) FIG. 12 is a perspective view of a joint formed by interlocking metal parts with an overmold, in accordance with the present invention; and,

(14) FIG. 13 is a perspective view of a joint formed by interlocking metal parts with an overmold, in accordance with aspects of the present invention.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(15) The following description of the preferred embodiment(s) is merely exemplary in nature and is in no way intended to limit the invention, its application, or uses.

(16) Referring to FIGS. 1-13 generally, there is provided a three dimensional (3D) reinforcement shape in combination with the benefit of an overmolding process that at least partially overmolds said reinforcement. The alignment, assembly and joining of the 3D reinforcement is done either before the overmolding or in the injection tool. The present invention meets predetermined requirements such as strength standards. A predetermined geometry is provided by the reinforcement adapted to be overmolded. Typically, there is provided at least one 3D dimensioned reinforcement, e.g., rib, that is operably connected to at least one second reinforcement, e.g., plate. Preferably, a first reinforcement is operably connected to a panel and a second reinforcement is operably connected to the first reinforcement. More preferably, there is provided at least one plate that is operably connected to a panel or other component and at least one rib that is operably connected to the plate. Most preferably a metal plate is operably connected to a large panel, e.g., inner panel of a liftgate in a hinge area or any other predetermined area of the panel, and a metal rib is operably connected to the plate creating an interlocked 3D geometry, which can be overmolded. While a metal plate is described, it is understood that the first and second reinforcements can be any predetermined profile and reinforcement material depending on the particular application without departure from the scope of the present invention.

(17) The interlocked 3D reinforcement is pre-assembled by a predetermined step (e.g., slotted together, mechanical lock, through hole, etc and any combinations) or assembled in the injection tool.

(18) According to an aspect of the present invention, a metal bracket reinforcement is fully overmolded on one or both sides of the panel. According to another aspect of the present invention, a metal bracket reinforcement is at least partially covered on one or both sides of the panel. According to another aspect of the present invention, a metal bracket reinforcement is only partially visible from the B-side of the panel (e.g., open at rib nuts, or more open to leave room for functional openings, etc). According to another aspect of the present invention, a metal bracket reinforcement is only partially visible from the B-side of the panel (e.g., open at rib nuts, or more open to leave room for functional openings, etc) so that it is not fully overmolded or covered on that one side, and is fully open or substantially open on the other side of the panel. Preferably, when overmolded to be fully covered, it is a show quality surface or Class-A surface with no read through issues.

(19) According to an aspect of the present invention plastic flows between a first reinforcement and second reinforcement (e.g., 3D stamping reinforcement, 3D reinforcement channel, etc) in predetermined locations, e.g., such as between a raised rib that has a channel). At least one aperture, e.g., bleeder hole, is provided, e.g., through backside of the second reinforcement, for delivery of plastic to predetermined locations. According to an aspect of the present invention, the plastic is on top of the 3D structure (e.g., on top of the 3D structure in addition to within the raised rib channel within the 3D structure). Any of the aforementioned aspects are contemplated in combination depending on the application without departure from the scope of the present invention.

(20) The 3D geometry of the present invention has significant advantages over flat plates, and the present invention has significant advantages over welding. One advantage is that the interlocking 3D geometry prevents parts from sliding over each other to break the welds.

(21) The reinforcements of the present invention are preferably metal, e.g., steel, aluminum, the same metal, or different metals, etc.

(22) Welding at least one metal rib on an existing metal bracket would have disadvantages. A full

line of welding on a complex geometry is difficult, time consuming and expensive, and spot welding is not strong enough. Thus, referring to the figures generally, according to aspects of the present invention, provide a new metal rib in accordance with the present invention and make the connection with a plastic material. Another advantage is that pre-assembly costs of material can be eliminated or greatly reduced. In addition, by having no straight surface to surface (e.g. flat surface to flat surface interface), instead the present invention creates a flange or other suitable profile for connection, there is realized significant benefits in strength.

(23) Referring to the figures generally, there is depicted various exemplary flange options. It is understood that any alternative suitable flange/rib/profile and combinations suitable for interlocking features depending on the particular application are contemplated without departure from the scope of the present invention.

(24) Referring more particularly to FIGS. 1-2, there is depicted a metal overmolding joint shown generally at **10** including at least one reinforcement **12**, and including at least one 3D reinforcement **14** including at least one geometric feature **16** (e.g., at least two ribs **16**, opposing ribs, etc). The at least one 3D reinforcement is typically a bracket, e.g., L-bracket or any other predetermined suitable shape, including the at least one geometric feature **16** adjacent an aperture **17** formed in the 3D reinforcement **14**. FIGS. 1-2 depict two exemplary ribs that are identical and located at opposite sides of the aperture **17**. Another aperture **19** is formed in the at least one reinforcement **12**. When the at least one reinforcement **12** and the at least one 3D reinforcement **14** are then positioned relative to each other for creating the desired joint indicated generally at **28**, the apertures **17** and **19** are generally aligned and allows a predetermined interlocking geometry, indicated generally at **18**, adapted for joining the reinforcements. A predetermined plastic overmold **20** flows through the apertures **17,19**, within the interlocking geometry portion **18** and over the geometric features **16**. This sandwiches the reinforcements together forming the joint. Plastic, e.g., polypropylene, is delivered to join the reinforcements. By way of example, the plastic forms a panel **21** that abuts against a bottom surface **23** of the at least one reinforcement **12**. The plastic also fills the interlocking geometry portion **18**, including abutting against at least one bottom surface **25** of the 3D reinforcement **14**. The plastic also at least partly overmolds **20** the geometric features **16**. Thus, a strong reinforcement joint is formed without the need for welding. FIGS. 1-2 depict one interlocking geometry region. It is understood that more or less overmolded connection portions **27** are contemplated depending on the application without departure from the scope of the present invention.

(25) The reinforcement **12** is preferably metal, e.g., aluminum, steel, etc. The 3D reinforcement **14** and geometric feature **16** are preferably metal, e.g., aluminum, steel, etc. The reinforcement **12** is preferably steel and the geometric feature **14** and bracket **16** are preferably aluminum.

(26) Referring more particularly to FIGS. 3-4, there is depicted a metal overmolding joint shown generally at **10** including at least one reinforcement **12**, and including at least one 3D reinforcement **14** including at least one geometric feature **16** (e.g., at least two flanges **16**, at least two legs, a plurality of opposite facing legs, at least two flanges facing one direction and at least one additional flange generally facing an opposite direction, ribs, etc). FIGS. 3-4 depict three exemplary geometric features **14** that extend in opposite directions on lower portions of the reinforcement **16** (two in one direction, and the other in a second direction). Another aperture **19** is formed in the at least one reinforcement **12**. The at least one 3D reinforcement is typically a bracket including the at least one geometric feature **16** adjacent an aperture **17** formed in the 3D reinforcement **14**. Another aperture **19** is formed in the at least one reinforcement **12**. When the at least one reinforcement **12** and the at least one 3D reinforcement **14** are then positioned relative to each other for creating the desired joint, the apertures **17** and **19** are generally aligned and allows a predetermined interlocking geometry, indicated generally at **18**, adapted for joining the reinforcements. A predetermined plastic overmold **20** flows through the apertures **17,19**, within the interlocking geometry portion **18** and over the geometric features **14**. This sandwiches the reinforcements together forming the joint.

Plastic, e.g., polypropylene, is delivered to join the reinforcements. By way of example, the plastic forms the panel **21** that abuts against a bottom surface **23** of the at least one reinforcement **12**. The plastic also fills the interlocking geometry portion **18**, including both apertures **17** and **19**. The plastic also at least partly overmolds **20** the geometric features **16**. Thus, a strong reinforcement joint is formed without the need for welding. FIG. **3** depicts three interlocking geometry regions. It is understood that more or less overmolded interlocking connections **27** are contemplated depending on the application without departure from the scope of the present invention.

(27) Referring more particularly to FIGS. **5-7**, there is depicted a metal overmolding joint shown generally at **10** including at least one reinforcement **12** and at least one second reinforcement **14** including at least one geometric feature **16** with predetermined interlocking features that are at least partially overmolded **20**. Pre-assembly of the metal reinforcements is done without welding, then is overmolded. The geometric features **16** are preferably tabs that are bent or tabs **24** that are not bent, e.g., slid through slots **22** and into place for mechanical lock. Overmold is performed at least partly about tabs **24** and are interlocked at at least one interlocking portion **18**. The at least one reinforcement **12** and/or the second reinforcement **14** has/have at least one slot **22** to receive a corresponding interlocking feature **24** (e.g., tab). The figures depict to tab/slot mechanical interlocks, however, more or less are contemplated depending on the application without departure from the scope of the present invention. Optionally, the second reinforcement **14** can also have additional tabs **26** that fit into apertures **28**, e.g., apertures formed on sides **30,30** of the reinforcement **12**. Any suitable arrangement and combination is contemplated depending on the particular application without departure from the scope of the present invention. The interlocked unit is then overmolded **20**. The slots on the base and slots in the cutouts interlock by sliding and/or bending to snap in.

(28) Referring more particularly to FIG. **8**, there is depicted a metal overmolding joint assembly shown generally at **10** including at least one reinforcement **12**, and at least one second reinforcement **14** with predetermined geometric features **16** for interlocking with the reinforcement **12**, e.g., at least one slot/tab **22/16** arrangement formed by the reinforcements. The geometric feature **16** is slid in place and bent or not bent, by way of example. Once the reinforcements **12/14** are interlocked, plastic is delivered and at least partially overmolds **20** the part. Preferably, at least one rib **24** is created with the overmold plastic, e.g., polypropylene, as well as a panel **21** positioned to abut against the bottom surface of the at least one reinforcement **12**. The ribs **24** are perform as a fixed position of reinforcement. Apertures can be provided in the metal reinforcement **14** adjacent the ribs **24** to at least partially overmold **20** the plastic against the second reinforcement **14**. Most preferably, the slot/tab **22/16** are interlocking metal to metal.

(29) Referring more particularly to FIG. **9**, there is depicted a metal rib overmolding joint shown generally at **10** including at least one reinforcement **12** with at least one 3D reinforcement **14** with at least one geometric feature **16**. The at least one second reinforcement is at least partially overmolded **20**. The reinforcement **12** preferably is against at least one surface of a panel **23** formed of plastic, e.g., polypropylene. The reinforcements are preferably metal, e.g., steel, aluminum, different metal, same metal, etc. FIG. **9** incorporates L-rib overmolds.

(30) Referring more particularly to FIG. **10**, there is depicted a metal rib overmolding joint shown generally at **10** including at least one reinforcement **12** with at least one other reinforcement **14**, preferably at least two reinforcements that are spaced apart **14,16** or a single piece **14** that has at least one aperture **17** punched through. The part is overmolded **20**. Molded ribs **22** can be formed at time of overmold. Molded supports **24** can be formed at time of overmold. At least one second aperture **19** is formed in the reinforcement **12**, preferably, generally in line with the second reinforcement **14**. When the reinforcements **12** and **14** are positioned relative to each other for creating the desired joint, the aperture **19** and reinforcement **14** are generally aligned and allows a predetermined interlocking geometry, indicated generally at **18**, adapted for joining the

reinforcements. Plastic, e.g., polypropylene, is delivered to join the reinforcements. By way of example, the plastic forms a panel **21** that abuts against a bottom surface **23** of the at least one reinforcement **12**, and the plastic overmold **20** flows through the aperture **19**, within the interlocking geometry portion **18**, and at least partially over the geometric features of the reinforcement **14**—securing the reinforcements **12,14** in place. Thus, a strong reinforcement joint is formed without the need for welding. FIG. **10** depicts one interlocking geometry region. It is understood that more or less overmolded connections are contemplated depending on the application without departure from the scope of the present invention.

(31) Referring more particularly to FIG. **11**, there is depicted a metal rib overmolding joint shown generally at **10** including at least one reinforcement **12**, and including at least one second reinforcement **14** with at least one geometric feature **16**. Plastic forms a panel **21** portion, e.g., polypropylene. The part is overmolded **20**. The reinforcements **12,14** are preferably both steel, aluminum, etc. An aperture can be added, preferably in the Y-part, to hide any sink mark in the radius/transition of the part. The plastic overmold **20** is located at least partially over the geometric features of the reinforcement **14**—securing the reinforcements **12,14** in place. Thus, a strong reinforcement joint is formed without the need for welding. FIG. **11** depicts one interlocking geometry region. It is understood that more or less overmolded connections are contemplated depending on the application without departure from the scope of the present invention.

(32) FIG. **12** is similar but a mechanical lock is added **40**, and an aperture **42** is included to shoot in plastic material, e.g., polypropylene. There is provided the mechanical lock **40** (e.g., not parallel, but a 3D shape) with an air pocket **44** in-between parts for the flow of plastic.

(33) Referring more particularly to FIG. **13**, there is depicted a metal rib overmolding joint shown generally at **10** including at least one reinforcement **12**, and at least one second reinforcement **14** with geometric features **16a,16b** (e.g., at least two flanges **16**, at least two legs, a plurality of opposite facing legs, at least two flanges facing one direction and at least one additional flange generally facing an opposite direction, ribs, etc). At least one aperture **19** is formed in the at least one reinforcement **12**, and the second reinforcement's **14** legs **16a,16b** are pushed through the aperture **19** and flange out on the other side of the reinforcement **12**. Preferably, there's a Y-shape cut in metal bracket **14**, and when it's pushed through the aperture **19**, it flanges out and then it becomes fixed with plastic overmold **20**. The panel **21** of plastic is against the under surface of the reinforcement **12** and surrounds the legs **16a,16b**.

(34) While an inner panel of a liftgate is shown, it is understood that the present invention is applicable to and adaptable to any vehicle panel, especially large panels, such as, but not limited to, door panels, door deck lids, enclosure panels, liftgates, tailgates, windshield panels, window panels, etc and any other panel without departure from the scope of the present invention.

(35) While a hinge reinforcement area is shown, it is understood that the present invention is applicable to and adaptable to any area of a large panel, such as, but not limited to, upper hinge area, strut area, ball joint area, windshield area, window area, license plate area, garnish trim, side trim, ribbing, etc. and any other area or on any other panel and area without departure from the scope of the present invention.

(36) While the plate/ribs are metal, preferably steel, it is understood that alternative materials suitable for meeting predetermined requirements depending on the application are contemplated without departure from the scope of the present invention.

(37) While “rib” is mentioned throughout, it is understood any suitable structure depending on the application is contemplated without departure from the scope of the present invention. While “plate” is mentioned throughout, it is understood any suitable structure depending on the application is contemplated without departure from the scope of the present invention.

(38) The description of the invention is merely exemplary in nature and, thus, variations that do not depart from the gist of the invention are intended to be within the scope of the invention. Such variations are not to be regarded as a departure from the spirit and scope of the invention.

Claims

1. A metal rib overmolding joint assembly operable for a liftgate, comprising: at least one first localized reinforcement that is a metal base plate and includes at least one predetermined geometric feature; at least one three-dimensional localized reinforcement that is metal and includes a plurality of geometric features positioned relative to the at least one geometric feature of the at least one first localized reinforcement, the plurality of geometric features operably positioned and secured to said at least one first localized reinforcement operably adapted for creating a reinforcement interlock without a weld, wherein said geometric features include a plurality of interlocking slots and tabs that interlock the first three-dimensional localized reinforcements, wherein the slots are provided on the first reinforcement to form the predetermined geometric feature, the tabs are provided on the three-dimensional reinforcement, the tabs are L-shaped, and the tabs slide through the slots and into place for mechanical lock to interlock the first reinforcement and the three-dimensional reinforcement without bending the tabs; and a three-dimensional overmold forming a joint overmolded connection for transferring load, wherein at least one side of the three-dimensional overmold is a portion of a liftgate panel, wherein said first and second localized reinforcements in combination form a cruciform-like shape encapsulated by said overmold.
2. A metal rib overmolding operably adaptable for a liftgate, comprising: at least one first localized reinforcement that is metal, said at least one first localized reinforcement including at least one predetermined geometric feature; at least one second localized reinforcement including at least one geometric feature operably adapted to join to said at least one first localized reinforcement at a joint without welding; wherein said geometric features include a plurality of interlocking slots and tabs that interlock the first and second localized reinforcements, wherein the slots are provided on the first reinforcement, and the tabs are provided on the second reinforcement, said tabs are L-shaped and inserted through said slots and slid into place and not bent for mechanical lock to interlock the first and second reinforcements; at least a partial overmold of said joint creating an overmolding connection to transfer load forces, wherein said first and second localized reinforcements together form a cruciform-like shape cross section encapsulated by said overmold; at least one panel, wherein at least portions of said first and second localized reinforcements are sandwiched between the panel and the overmold and at least one side of the overmold is a vehicle panel.
3. The metal rib overmolding of claim 2, wherein at least one of said first and second localized reinforcements is an upstanding three-dimensional feature.
4. The metal rib overmolding of claim 2, wherein at least one of said first and second localized reinforcements is a three-dimensional reinforcement including a plurality of ribs.
5. The metal rib overmolding of claim 2, further comprising a plurality of ribs formed of the overmold material in a fixed position for additional reinforcement.
6. The metal rib overmolding of claim 2, further including a plurality of substantially L-shaped rib overmolds engaging the first and second localized reinforcements.
7. The metal rib overmolding of claim 2, wherein the first and second localized reinforcements are metal, interlocking metal to metal.
8. The metal rib overmolding of claim 2, wherein at least one reinforcement interlock includes delivered plastic material to at least an interlocking geometry portion providing a joint and overmolding the joint.
9. The metal rib overmolding of claim 2, further comprising additional interlocking apertures/tabs that interlock the first and second localized reinforcements.
10. The metal rib overmolding of claim 9, wherein the interlocking geometry at said joint is overmolded by polypropylene.
11. The metal rib overmolding of claim 2, wherein plastic flows between the first localized reinforcement and second localized reinforcement in predetermined locations via at least one

bleeder hole to deliver plastic to the predetermined locations.

12. The metal rib overmolding of claim 11, wherein the plastic is over the at least one geometric feature and within a raised rib channel within a three-dimensional structure.

13. A method of manufacturing a metal reinforcement joint operable for a liftgate, comprising: providing an injection molding tool; providing at least one first localized reinforcement with predetermined geometric features; providing at least one second localized reinforcement with predetermined geometric features operable to secure said at least one first and second localized reinforcements together; positioning said at least one first and second localized reinforcements relative to each other; wherein said geometric features include a plurality of interlocking slots and tabs that interlock the first and second localized reinforcements, wherein the slots are provided on the first reinforcement, and the tabs are provided on the second reinforcement to form the geometric feature; joining said at least one first and second localized reinforcements forming at least one joint portion, wherein the tabs are L-shaped and inserted through the slots and slid into place for mechanical lock to interlock the first and second reinforcements without bending the tabs, wherein when joined, a first portion of said second localized reinforcement contacts said first localized reinforcement on a first plane and a second portion of said second localized reinforcement extends from said first portion on a second plane and angle oblique to said first plane; overmolding at least said at least one joint portion creating a load transferring overmolding connection, wherein said first and second localized reinforcements in combination form a cruciform-like shape cross section encapsulated by said overmold, and wherein at least one side of the overmolding forms a portion of a liftgate panel.

14. The method of claim 13, wherein one of the at least one first and second localized reinforcements is at least one three-dimensional metal rib, and the other one of said at least one first and second localized reinforcements is a metal base plate.

15. The method of claim 13, further comprising slotting said at least one first localized reinforcement to join the first and second localized reinforcements.

16. The method of claim 13, wherein joining further comprises pre-assembling the interlocking parts joint or assembling in the injection molding tool.

17. The method of claim 13, wherein said joining of said at least one first and second localized reinforcement is done by interconnecting the respective first and second localized reinforcements together forming a joint to be overmolded in the injection molding tool.

18. The method of claim 13, wherein said joining of said at least one first and second localized reinforcement is pre-formed and then placed in the injection molded tool.

19. The method of claim 13, wherein the at least one first and localized second reinforcements are steel or aluminum.

20. The method of claim 13, wherein overmolding is three dimensional overmolding of at least the joint that is mechanically securing the at least one first and second localized reinforcements together.

21. The method of claim 13, wherein the at least one first and second localized reinforcements create joints without welding.

22. The metal rib overmolding claim 13, wherein the reinforcement interlock is pre-assembled outside of an injection molding tool, interlocking the first and second localized reinforcements without welding.

23. The metal rib overmolding claim 13, wherein the reinforcement interlock is assembled in an injection molding tool before overmolding and without metal welding.

24. The metal rib overmolding claim 13, further comprising providing an injection tool with active support features.

25. The metal rib overmolding claim 13, further comprising providing an injection tool with a lifter injector pin selectively cycling in/out.
